

A Sensor-to-Brake AEB Pipeline in CARLA with Object-Level Radar, Camera Verification, and Limit-Finding Evaluation

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Abstract—An automatic emergency braking (AEB) simulation can appear successful if it is given a perfect lead-vehicle state; the harder problem is to construct a stable, collision-relevant target from sensor outputs and close the loop through vehicle actuation. This paper presents a sensor-to-brake pipeline in CARLA 0.9.11. Sparse forward-radar detections are filtered against a predicted ego-path corridor, clustered, and tracked before target selection. Radar retains ownership of range, relative velocity, time-to-collision (TTC), and stopping-distance margin. A YOLO26n detector provides a late semantic permission gate: a provisional brake is applied only when the selected radar target projects into a current or recent camera car box. A staged PID law then maps risk to the CARLA brake command. The evaluation comprises 66 single-run, fixed-tick simulations: 24 moving-lead, 30 braking-lead, and 12 cut-in cases. All 38 intended-range cases pass; 25 of 28 stress cases pass. The retained collisions occur for braking leads at 95 and 110 km/h with a 20-m gap and for a 100/60-km/h late cut-in. These outcomes map a configured operating boundary rather than estimate safety reliability. The work is a transparent engineering baseline, not a claim of a new fusion primitive, NCAP certification, or real-vehicle validation.

Index Terms—automatic emergency braking, CARLA, radar-camera verification, target selection, time-to-collision, vehicle control

I. INTRODUCTION

AEB is a closed-loop perception–decision–actuation problem. Radar can measure range and radial relative velocity, but one return need not represent a vehicle in the ego path. A camera can identify vehicles, but does not by itself provide the same direct closing-speed measurement. A braking function must resolve both questions before the remaining distance is exhausted. AEB reviews consequently treat environment perception, decision logic, and control execution as an interdependent chain [1].

High-fidelity simulation supports repeatable near-collision experiments before track testing [2], [3]. It also creates a methodological trap: a simulator exposes actor identity, exact object motion, and ground-truth boxes. Feeding those quantities to the online AEB logic tests a controller with ideal target state rather than a sensor-to-brake system. This study therefore begins with CARLA radar detections and RGB frames. Actor identity and actor kinematics are excluded from the brake decision, although simulator geometry is used for calibrated projection and road-height rejection. Ground truth is reserved for offline labels and outcome scoring.

None of the individual building blocks—geometric association, TTC, stopping distance, path relevance, or staged braking—is claimed as new. The contribution is a traceable configuration of those blocks and an evaluation that preserves its failures. Specifically, the paper provides: 1) a point-to-object layer for CARLA radar with path filtering, deterministic clustering, and temporal confirmation; 2) a radar-derived risk chain with a late YOLO semantic permission gate, without online actor IDs or ground-truth vehicle state; 3) closed-loop staged-PID actuation with fixed-tick logging; and 4) a parameterized evaluation that separates an intended speed range from limit-finding cases and reports all three collisions.

II. RELATED WORK AND POSITIONING

Radar–camera AEB predates this study. Kim and Song associated radar and vision for vehicle recognition [4], while Hsu *et al.* addressed radar/camera noise filtering for AEB target selection [5]. Geometric in-path association has also been demonstrated with LiDAR and camera on a real vehicle [6]. For decision and control, Zhang *et al.* combined curved-road target recognition, TTC, safety distance, and graded braking in CarSim/Simulink [7]. Thus neither path gating nor multi-criterion staged braking is a first contribution here.

CARLA has also been used to validate full autonomous-driving pipelines under protocol-derived scenarios [8]. More specifically for AEB, Feng *et al.* evaluated event-camera perception in CARLA [9]. Akula *et al.* implemented a radar-guided camera verifier on an instrumented vehicle, replacing full-frame recognition with an edge-density test inside radar regions [10]. The present camera role is philosophically similar, but uses YOLO for semantic *car* confirmation. Its distinct focus is the CARLA-specific conversion of sparse radar points into tracks, path-aware kinematic risk, staged-PID actuation, and a 66-case car-to-car boundary sweep. Table I summarizes this positioning. It supports a system integration claim, not algorithmic firstness.

III. EVALUATED SENSOR-TO-BRAKE PIPELINE

A. Scope and information boundary

The implementation uses a Tesla Model 3 in Town04 under ideal weather. A windshield RGB camera and bumper radar run at 20 Hz. CARLA radar is an abstract detection sensor,

TABLE I
POSITIONING AGAINST REPRESENTATIVE AEB AND IN-PATH FUSION STUDIES.

Study	Perception / target role	Decision or actuation	Evaluation emphasis
Kim and Song [4] Bae <i>et al.</i> [6]	Radar-vision vehicle recognition LiDAR projection and camera tracking for closest in-path vehicle	AEB-oriented target recognition Euro NCAP-inspired AEB use	Sensor-fusion recognition Real-vehicle scenarios
Zhang <i>et al.</i> [7]	Curved-road hazardous-target selection	TTC/safety-distance fusion and graded braking	CarSim/Simulink
Feng <i>et al.</i> [9]	Event-camera detection compared with RGB	Earlier visual detection for AEB	CARLA DVS scenarios
Akula <i>et al.</i> [10]	Automotive-radar tracks and edge- density camera verification	TTC trigger and brake-by-wire	72 real driving sessions; 33 staged threats
This work	CARLA point-to-track radar; path gate; YOLO semantic permission	TTC, stopping margin, staged PID, CARLA actuation	66 fixed-tick moving-lead, braking-lead, and cut-in cases

not a raw FMCW signal chain; virtual radar fidelity can differ substantially by model level [11]. The paper therefore claims logic validation only, not radar signal-processing validation.

Fig. 1 states the online information boundary. Exact sensor transforms calibrate radar-to-image projection, and a CARLA map waypoint query estimates local road height for ground rejection. These are privileged simulator geometries and are disclosed explicitly. Online braking does not query actor ID, actor speed, actor lane label, or ground-truth 2-D/3-D boxes. The latter are used offline to create YOLO labels and to score collision, gap, and target role.

B. Online execution order

The ordering of the blocks matters because camera confirmation is not allowed to create kinematic risk on its own. At every radar update, the implementation: (1) samples ego motion and updates the constant-curvature path; (2) rejects radar returns outside the range, height, road-ground, and path gates; (3) clusters the remaining returns and associates them with existing tracks; (4) forms a `RadarObject` list and selects one confirmed target; (5) computes TTC, required stopping distance, supervisory state, and a provisional brake value; (6) runs YOLO on the latest available RGB frame and projects the selected radar target into that image; (7) permits a provisional BRAKE only if the image association is current or held; and (8) applies `VehicleControl` and logs the resulting vehicle state at the next synchronous tick.

This ordering encodes explicit information ownership. A YOLO box without a radar target cannot command braking because it supplies neither radial velocity nor the distance used by the risk equations. Conversely, a closing radar object can reach WARNING, but a provisional BRAKE is vetoed if it lacks recent visual association. The design is intentionally conservative about radar false positives and intentionally vulnerable to a camera miss; this asymmetry is revisited in the cut-in result. Inference is configured every 0.15 s, slower than the 0.05-s sensor tick, which makes the short confirmation hold part of the evaluated behavior rather than a cosmetic display feature.

C. Radar objects, path relevance, and target selection

A constant-curvature path is estimated from ego speed, yaw rate, and steering. For curvature κ and arc length s ,

$$x(s) = \frac{\sin(\kappa s)}{\kappa}, \quad y(s) = \frac{1 - \cos(\kappa s)}{\kappa}, \quad (1)$$

with the straight-line limit used near $\kappa = 0$. The horizon is capped by the 100-m radar range and by a 2.5-s speed look-ahead. Returns outside configured range, height, road-ground, or 1.25-m path-distance gates are rejected.

Remaining returns are connected when planar separation, vertical separation, and radial-velocity difference are within 1.0 m, 1.5 m, and 2.0 m/s. A cluster's longitudinal coordinate is its 20th percentile; lateral coordinate, height, and relative velocity are medians. Greedy association compares predicted position and velocity across frames. A track requires three consecutive hits and is stale on a miss. Confirmed, non-stale tracks are ranked by finite TTC and then range. The selected track must persist for five frames unless its longitudinal distance is at most 22 m or its stopping margin is at most -4 m, in which case the risk bypass avoids waiting for all five selections. An unmatched track is immediately marked stale and cannot be selected; it is deleted after four missed updates. Because the point clusters are small, connected components are built by a direct pairwise neighborhood test and tracks are matched greedily by normalized position and velocity error. These choices favor inspectability over optimal data association. They are deterministic engineering rules, not a Kalman filter, a probabilistic multi-object tracker, or a calibrated existence score.

The two temporal gates serve different purposes. Three radar hits establish that a cluster is persistent; five repeated target selections establish that the same confirmed track remains the highest-risk candidate. The emergency bypass prevents these delays from being unconditional. At 20 Hz, a clean new cluster is confirmed on its third sample (0.10 s after the first); because target counting starts then, it reaches five selections on the seventh sample (0.30 s after the first), absent bypass. Association and sensor arrival can change the exact first usable tick.

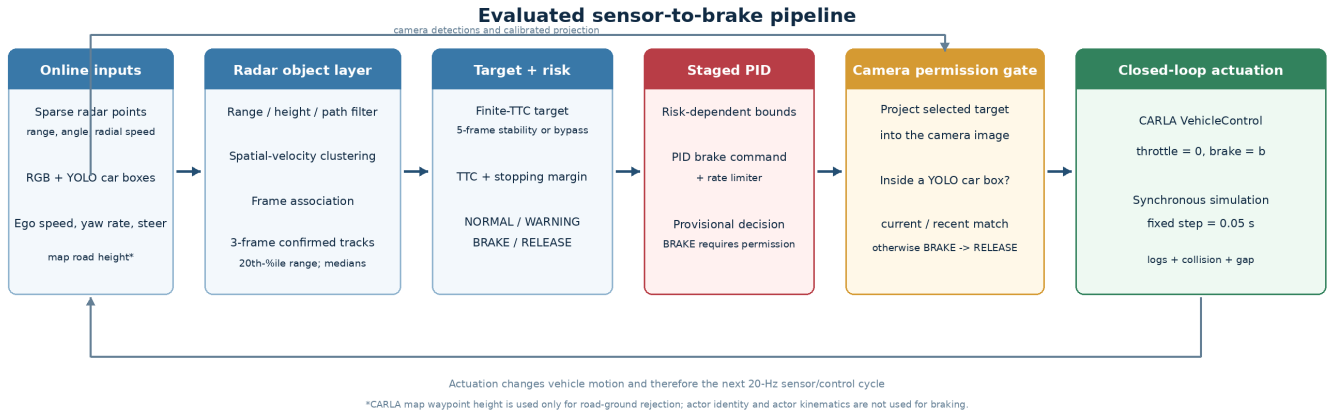


Fig. 1. Evaluated closed loop. Radar creates the kinematic target and a provisional AEB decision; camera association is a permission gate on BRAKE. The asterisk marks the disclosed map-height query.

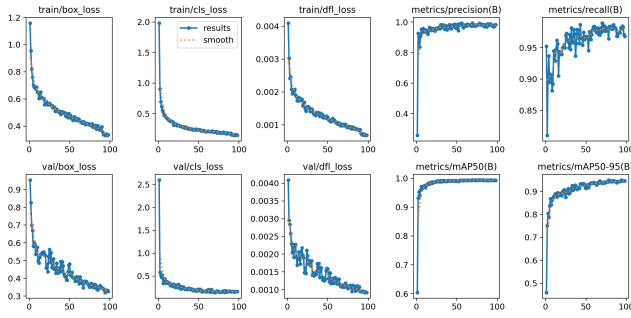


Fig. 2. Training and validation history of the one-class YOLO26n verifier. Detector metrics are not treated as AEB safety metrics.

D. Camera verification

YOLO26n [12] was fine-tuned for one *car* class using CARLA-generated labels. For collection only, visible 3-D actor boxes are projected into the runtime camera view and converted to YOLO labels. Sessions are separated across train, validation, and test splits; 16–21% empty images are retained to expose the detector to negative frames. The online AEB never receives these labels.

The splits contain 1,505/300/200 images and 1,872/379/264 boxes. Training uses 640×640 images, 100 epochs, batch size 16, AdamW, initial learning rate 0.001, and seed 2026. At the reported best epoch, precision, recall, mAP₅₀, and mAP_{50:95} are 0.9919, 0.9642, 0.9940, and 0.9495. Fig. 2 shows the checked-in training history. These values establish narrow in-domain suitability; they do not measure end-to-end association accuracy or transfer beyond Town04 and favorable imaging conditions.

For a target in camera coordinates (X_c, Y_c, Z_c) ,

$$u = f_x X_c / Z_c + c_x, \quad v = f_y Y_c / Z_c + c_y, \quad Z_c > 0. \quad (2)$$

A match occurs when (u, v) lies inside a non-maximum-suppressed YOLO car box. Only a provisional BRAKE is gated: a current or recently held match passes it; otherwise the command becomes RELEASE with zero brake. The evaluated

implementation holds confirmation for 0.35 s using the host monotonic clock. Radar remains the sole source of range and relative velocity. This is late decision gating, not joint multi-object fusion.

E. Risk and staged-PID actuation

With longitudinal distance d and radar relative velocity v_r (negative when closing),

$$v_c = -v_r, \quad \text{TTC} = d/v_c \quad (v_c > 0). \quad (3)$$

For ego speed v_e and $v_t = \max(0, v_e + v_r)$, the required distance is

$$d_{\text{req}} = \max\left(d_0, v_e t_r + \frac{v_e^2}{2a_e} - \frac{v_t^2}{2a_t} + d_0\right), \quad m = d - d_{\text{req}}. \quad (4)$$

Here $t_r = 0.20$ s, $a_e = 8.0$ m/s², $a_t = 6.0$ m/s², and $d_0 = 1.0$ m are risk-model parameters, not measured hardware properties.

The supervisory states are NORMAL, WARNING, BRAKE, and RELEASE. Inside BRAKE, the PID error combines margin and TTC deficits:

$$e = \max(0, m_{\text{th}} - m - \delta_m) + k_t \max(0, T_b - \text{TTC}), \quad (5)$$

$$b = \text{clip}\left(b_0 + K_p e + K_i \int e dt + K_d \dot{e}\right).$$

The output is rate-limited and clipped by soft, medium, hard, or emergency risk bounds (0.55, 0.75, 0.90, and 1.00). A distance of at most 18 m, margin at most −5 m, or TTC at most 0.80 s opens the emergency schedule. Hard braking is available at TTC at most 1.10 s or margin at most −2 m. Outside those regions, PID operates within medium or soft bounds. The integral is clamped, only a positive error derivative is retained, and the command rise/fall rates are limited. The state logic includes a minimum 0.3-s hold and a 3.5-s release TTC. In the evaluated batch, `hold_brake_until_stopped` is enabled, so the provisional controller remains latched through the stopping phase unless the camera permission gate blocks it; control state is reset between scenarios.

TABLE II
PRINCIPAL EVALUATED CONFIGURATION.

Item	Value
Camera / FOV / tick	1280×720 / 70° / 0.05 s
Radar range / FOV / rate	100 m / 30°×6° / 2000 points/s
Track / selected-target confirmation	3 / 5 frames
Warning / brake / release TTC	3.0 / 1.5 / 3.5 s
PID K_p, K_i, K_d, k_t	0.12, 0.01, 0, 0.12
Brake rise / emergency rise	3.0 / 20.0 s ⁻¹
Hard / emergency TTC	1.10 / 0.80 s
Hard / emergency margin	-2.0 / -5.0 m

Thus stage labels are command schedules inside BRAKE, not separate supervisory states. The design is also not a comfort-optimal longitudinal controller: $K_d = 0$ in the final configuration, and no acceleration/jerk constraint is part of the objective. Staging is used to make risk-dependent bounds explicit while retaining a continuous command. Table II reports the principal reproducibility parameters.

IV. EXPERIMENTAL DESIGN

The simulator runs synchronously with a 0.05-s fixed step and physics-based throttle/brake control. The final suite contains only hazardous cases and every case expects braking without collision. PASS requires all configured checks: (1) a brake override occurs, (2) CARLA reports no collision, (3) minimum bumper gap is at least 0.5 m, and (4) maximum lane-center offset is at most 1.25 m. Clear-road, adjacent-lane, curve, cut-out, and multi-actor runs were development checks, but are not included in the 66-case aggregate and cannot support a quantitative false-brake claim here.

All cases use the same Town04 spawn region and physics speed controller. CCRm holds a slower target at 30 km/h below ego speed. CCRb begins with equal ego and lead speeds; the lead applies full braking 1.5 s after scenario start. Cut-in spawns the hazard in the left adjacent lane, starts a right lane change at 1.0 s, and requires that the selected brake actor be the cut-in vehicle. Thus the three families isolate persistent closing speed, a change in lead acceleration, and a change in lateral path relevance. They do not reproduce Euro NCAP apparatus or all protocol tolerances.

At every tick, the logger records simulator time, ego speed/acceleration, lane offset, collision count, center and bumper distances to the configured hazard, radar candidate/cluster counts, selected track, TTC, required distance, margin, AEB state, and brake command. Ground-truth actor matching appears only in the logger to check whether the selected target corresponds to the configured hazard. Quantitative timing uses simulator ticks rather than video FPS; the UI and video are diagnostic views only.

A. Measurement semantics and traceability

Two distance quantities must not be conflated. The AEB controller consumes the selected radar object’s longitudinal coordinate. By contrast, reported *bumper gap* is an evaluation

TABLE III
FINAL 66-CASE SUITE. EACH LISTED SPEED SET IS CROSSED WITH ALL GAPS.

Family	Intended range	Stress range
CCRM	ego/lead 60/30, 80/50; gaps 20,30,40,50,60,80 m (12)	100/70, 110/80; same gaps (12)
CCRB	ego 50,65,80; gaps 20,30,40,50,60,80 m (18)	ego 95,110; same gaps (12)
Cut-in	ego/lead 60/40, 80/50; gaps 25,35,45,60 m (8)	100/60; same gaps (4)
Total	38 cases	28 cases

TABLE IV
OUTCOME COUNTS BY FAMILY AND EVALUATION SUBSET.

Family	Intended			Stress		
	N	P	F	N	P	F
CCRM	12	12	0	12	12	0
CCRB	18	18	0	12	10	2
Cut-in	8	8	0	4	3	1
Total	38	38	0	28	25	3

quantity: the scenario runner projects the hazard center onto the ego forward axis and subtracts both actors’ bounding-box half-lengths. “Brake gap” is that evaluation gap at the first tick with an active AEB override, and “minimum gap” is its minimum over the run. CARLA’s collision event is an independent scoring signal. Consequently, evaluation uses privileged truth to judge a sensor-based decision, not to produce it.

The evaluated configuration is centralized in `configs/sensors.yaml`; the 66 scenarios are enumerated in `configs/scenarios/suites/system_limit_extended_sweep.yaml`. `radar_object_tracker.py`, `radar_aeb_pipeline.py`, `brake.py`, and `run_fusion_aeb_scenarios.py` implement the four principal stages. The run script writes per-tick CSV, per-case summaries, and aggregate JSON. This module/configuration mapping is included because a parameter table alone cannot specify execution order, bypasses, or stale-track behavior.

The ODD is narrowly defined as Town04 highway driving, cars only, ideal weather, and the stated simulated sensors. Within that ODD, Table III divides speed combinations into an *intended-range subset* and a *stress subset*; the subsets are not themselves ODDs. Gaps are swept for each speed pair. Each configuration is run once, so outcomes measure configured scenario coverage, not a probability of safe operation.

V. RESULTS AND DISCUSSION

A. Configured-case outcomes

Table IV reports 38/38 intended-range and 25/28 stress outcomes. CCRM passes all 24 cases because its 30-km/h closing speed is moderate even at high absolute ego speed. The failures are confined to CCRb at 95 and 110 km/h with a 20-m gap, and cut-in at 100/60 km/h with a 25-m gap.

The CCRb grid in Table V makes the limit localized rather than suggesting that all high-speed cases fail. Both collisions

TABLE V
CCRB OUTCOME MAP (P: PASS; C: COLLISION).

Ego speed	20 m	30 m	40 m	50 m	60 m	80 m
50 km/h	P	P	P	P	P	P
65 km/h	P	P	P	P	P	P
80 km/h	P	P	P	P	P	P
95 km/h	C	P	P	P	P	P
110 km/h	C	P	P	P	P	P

TABLE VI
REPRESENTATIVE BOUNDARY CASES FROM THE CHECKED-IN REPORT
AND EVIDENCE SUMMARY.

Scenario	Brake gap (m)	Min. gap (m)	Outcome
ccrm_110_80_gap_20	18.96	15.01	Pass
ccrb_80_gap_20	18.56	1.96	Pass
ccrb_95_gap_20	18.56	0.0134	Collision
ccrb_110_gap_20	—	0.0806	Collision
cutin_80_50_gap_25	10.46	5.51	Pass
cutin_100_60_gap_25	7.26	0.4763	Collision

occur only in the 20-m column; every configured gap from 30 to 80 m passes. Likewise, the 100/60-km/h cut-in fails at 25 m but passes at 35, 45, and 60 m. The evidence therefore describes a speed–gap–maneuver interaction, not a universal speed threshold.

Representative logs in Table VI separate absolute speed from closing dynamics. The 110/80-km/h CCRm case starts braking at 18.96 m yet retains 15.01 m, whereas a braking lead removes that favorable target motion. At the same 20-m initial gap, CCRb passes at 80 km/h with 1.96 m remaining but collides at 95 and 110 km/h. A brake command is present in the 95-km/h failure (Fig. 3); the evidence therefore supports an insufficient-distance boundary for the configured chain, not a claim that perception missed the lead.

B. Late cut-in boundary

For cut-in, risk becomes actionable only after the target enters the path corridor and receives usable radar/camera confirmation. The 80/50-km/h case brakes at 10.46 m and passes; the 100/60-km/h case brakes at 7.26 m and collides (Fig. 4). This exposes a trade-off rather than proving a component benefit: widening the corridor or bypassing camera permission might trigger earlier, but could also increase response to adjacent traffic. A matched radar-only/camera-gated ablation is required before assigning causality.

C. What the evidence establishes

Three conclusions are supported at different strengths. First, the full sensor-to-brake implementation can complete every configured intended-range case under one fixed CARLA setup. This is an end-to-end existence result: radar points, detector output, target gates, risk logic, and actuation all participate. Second, the parameter sweep identifies structured boundaries in the recorded run set: the CCRb failures share the shortest gap, and the cut-in failure combines the highest tested speed pair with the shortest cut-in gap. Third, logged first-brake

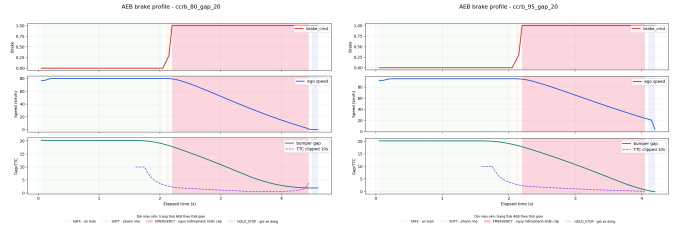


Fig. 3. CCRb at a 20-m initial gap: pass at 80 km/h (left) and collision at 95 km/h (right).

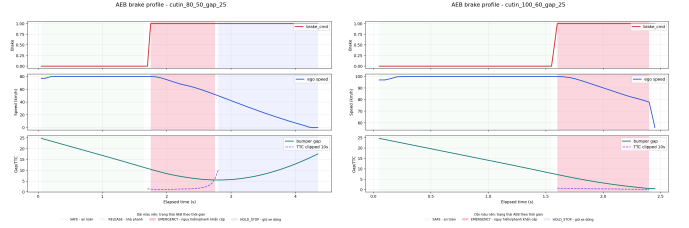


Fig. 4. Cut-in at a 25-m initial gap: pass at 80/50 km/h (left) and collision at 100/60 km/h (right).

distances show that target motion and path-entry time matter alongside absolute ego speed.

The same evidence does not isolate which module creates those outcomes. There is no run matrix that independently removes radar track confirmation, the predicted path, camera permission, stopping margin, or staged PID. It also contains no matched binary-brake, radar-only, or perfect-perception baseline under the final 66 configurations. Therefore wording such as “fusion improves safety,” “PID reduces jerk,” or “the path gate reduces false braking” would exceed the experiment. The defensible result is behavior of the integrated configuration.

D. Relation to the closest systems

The outcome should also be read relative to Table I. Compared with the real-vehicle radar-guided verifier of Akula *et al.* [10], this study has weaker external validity and no latency advantage to claim. Its camera stage is semantically stricter—it asks whether the target is a *car*, not merely whether edges occupy a radar-scoped region—but pays the cost of a learned model and in-domain data. Compared with Zhang *et al.* [7], it exposes the sensor-to-target construction rather than beginning from an already recognized hazard, but its controller has no vehicle-hardware calibration. Compared with the CARLA DVS work [9], it studies complementary radar-derived kinematics rather than visual update rate. These differences motivate the baseline; they do not establish superiority because datasets, platforms, and criteria differ.

E. Engineering implications

The retained failures suggest two concrete next experiments. For CCRb, speeds and gaps should be sampled more densely around 80–95 km/h and 20–30 m while recording target deceleration and brake onset relative to the 1.5-s event. This would distinguish control saturation from temporal perception

delay. For cut-in, a factorial comparison should vary corridor width, target-confirmation frames, and camera permission while measuring both collision and false-brake outcomes on paired adjacent-lane cases. Lateral velocity and time-to-lane-entry could then be added only if the baseline comparison shows that path-entry timing is the limiting factor.

VI. THREATS TO VALIDITY

A. Internal and construct validity

The risk thresholds, cluster gates, temporal confirmation, and PID gains were tuned within the same project before the final sweep. No held-out scenario optimization protocol is documented, so the 38/38 intended-range result may partly reflect configuration knowledge. PASS is also a binary construct: a 0.51-m minimum gap passes while a collision fails, but this does not grade severity, comfort, intervention appropriateness, or residual collision speed. The final suite contains hazards only, so it measures missed/late intervention more directly than unnecessary intervention.

The suite has one run per configuration and no randomized seeds, confidence intervals, or matched ablation. Consequently, 63/66 is a configured-case count, not a reliability estimate. The checked-in repository contains the final evidence summary and plots but not the referenced raw final CSV/JSON artifacts; reported aggregate and representative values therefore cannot be independently recomputed from this checkout alone.

CARLA radar supplies abstract detections rather than ADC samples, range-Doppler maps, CFAR, clutter, or production tracking. Camera labels, calibration, map height, and weather are simulator-domain quantities. The 0.35-s image-match hold uses host wall time rather than simulation time, so compute load can affect its simulated duration. No end-to-end latency distribution was measured. The tests exclude repeated trials, adverse weather, pedestrians, two-wheelers, tire-road variation, hydraulic delay, detailed ABS, and real hardware. Euro NCAP motivates scenario families only [13]; ISO 15623 motivates collision-warning quantities [14]. Neither constitutes certification.

B. External and reproducibility validity

All detector images and final scenarios share a narrow simulator domain. High in-domain mAP can coexist with brittle association under another map, vehicle shape, occlusion pattern, illumination, or camera calibration. Moreover, the final evidence pack records paths to per-tick CSV/JSON files that are absent from this checkout. The checked-in report, plots, configuration, and evidence summary support manuscript traceability, but not full third-party rerun or re-computation without CARLA 0.9.11, model artifacts, and the omitted logs. A release-quality replication package should pin the CARLA image, Python environments, model hash, random seeds, source commit, and complete generated summaries.

VII. CONCLUSION

This paper documented a CARLA sensor-to-brake AEB baseline in which sparse radar returns become confirmed, path-relevant objects; YOLO supplies late semantic permission;

and TTC, stopping margin, and staged PID close the control loop. All 38 intended-range configurations pass, while three of 28 stress configurations collide. The failures locate high-speed short-gap and late cut-in boundaries instead of being removed from the result. The next defensible step is repeated, seed-controlled radar-only versus camera-gated ablation with simulation-time synchronization and complete raw artifacts, followed by more realistic sensor and actuator models before any road-safety claim.

REFERENCES

- [1] L. Yang, Y. Yang, G. Wu, X. Zhao, S. Fang, X. Liao, R. Wang, and M. Zhang, "A systematic review of autonomous emergency braking system: Impact factor, technology, and performance evaluation," *Journal of Advanced Transportation*, vol. 2022, pp. 1–13, 2022.
- [2] A. Dosovitskiy, G. Ros, F. Codevilla, A. López, and V. Koltun, "CARLA: An open urban driving simulator," in *Proceedings of the 1st Conference on Robot Learning*, 2017, pp. 1–16.
- [3] Y. Kang, H. Yin, and C. Berger, "Test your self-driving algorithm: An overview of publicly available driving datasets and virtual testing environments," *IEEE Transactions on Intelligent Vehicles*, vol. 4, no. 2, pp. 171–185, 2019.
- [4] H. Tae Kim and B. Song, "Vehicle recognition based on radar and vision sensor fusion for automatic emergency braking," in *2013 13th International Conference on Control, Automation and Systems*, 2013, pp. 1342–1346.
- [5] P.-M. Hsu, M. H. Li, and K.-C. Chang, "Noise filtering in autonomous emergency braking systems with sensor fusions," *SAE International*, Tech. Rep. 2015-01-0216, 2015.
- [6] H. Bae, G. Lee, J. Yang, G. Shin, G. Choi, and Y. Lim, "Estimation of the closest in-path vehicle by low-channel LiDAR and camera sensor fusion for autonomous vehicles," *Sensors*, vol. 21, no. 9, p. 3124, 2021.
- [7] L. Zhang, Z. Yu, X. Xu, and Y. Yan, "Research on automatic emergency braking system based on target recognition and fusion control strategy in curved road," *Electronics*, vol. 12, no. 16, p. 3490, 2023.
- [8] C. Gómez-Huélamo, J. D. Egido, L. M. Bergasa, R. Barea, E. López-Guillén, F. Arango, J. Araluce, and J. López, "Train here, drive there: ROS-based end-to-end autonomous-driving pipeline validation in CARLA simulator using the NHTSA typology," *Multimedia Tools and Applications*, vol. 81, no. 3, pp. 4213–4240, 2022.
- [9] J. Feng, P. Zhao, J. Konpang, A. Sirikham, H. Zheng, P. Kalnaowakul, and J. Wang, "Validating DVS application in autonomous driving with various AEB scenarios in CARLA simulator," *World Electric Vehicle Journal*, vol. 16, no. 11, p. 634, 2025.
- [10] R. C. Akula, S. Kandhasamy, and M. Ganesan, "Radar-guided camera verification for automatic emergency braking: Rethinking object detection in radar-camera fusion," *arXiv preprint arXiv:2606.27556*, 2026. [Online]. Available: <https://arxiv.org/abs/2606.27556>
- [11] Z. F. Magosi, H. Li, P. Rosenberger, L. Wan, and A. Eichberger, "A survey on modelling of automotive radar sensors for virtual test and validation of automated driving," *Sensors*, vol. 22, no. 15, p. 5693, 2022.
- [12] Ultralytics, "Ultralytics YOLO documentation," [Online], accessed: 2026-08-18. [Online]. Available: <https://docs.ultralytics.com/>
- [13] Euro NCAP, "AEB car-to-car test protocol," Version 4.2, 2023.
- [14] *Transport Information and Control Systems—Forward Vehicle Collision Warning Systems—Performance Requirements and Test Procedures*, International Organization for Standardization Std. ISO 15 623:2013, 2013.